

Ananya Lahari Vadlamani

Robotics Engineering Student | Embedded Systems | Autonomous Systems

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PROFESSIONAL SUMMARY

Final-year Computer and Autonomous Systems Engineering student with High Distinction (HD) standing who loves building machines that sense, think, and act. Work experience spans robotics, embedded systems, and AI, from Python kinematics pipelines with robotic simulations to real-time Teensy 4.1 and ESP32 firmware for an award-winning knee exoskeleton and a smart safety walker. I enjoy taking an idea from simulation to working hardware: integrating sensors and actuators, writing safety-critical control logic, deploying edge-AI models, and testing until the system is reliable. I'm curious, hands-on, and motivated to build intelligent, connected, autonomous systems that make a real difference.

EDUCATION

University of Wollongong in Dubai (UOWD), Dubai, UAE

2023 – 2027

Bachelor of Engineering (Honours), Computers and Autonomous Systems Engineering

- Academic standing: High Distinction (HD) | Final-year student | Dean's Merit List Award for Academic Excellence, 2024
- Relevant coursework and project work: ROS2, FPGA and digital systems (Verilog/VHDL), embedded systems, control systems, robotics
- Bachelor's Thesis: working in robotics as part of the B.E. (Honours) program

St. Joseph's School, Abu Dhabi, UAE (CBSE)

2020 – 2023

Class XII: 95.6% (2022 – 2023) | Class X: 94.8% (2020 – 2021)

TECHNICAL SKILLS

- **Programming:** Python, C++, C, MATLAB, Java, SQL; working knowledge of C#, VHDL
- **Robotics and Simulation:** Robotic kinematics, Forward Kinematics (FK), Inverse Kinematics (IK), Jacobians, workspace reachability, singularity analysis, URDF, ROS2, MuJoCo, Pinocchio, Gazebo, Webots, SolidWorks
- **Embedded Systems:** Teensy 4.1, ESP32, Arduino, Raspberry Pi, Embedded C, PWM, ADC/DAC, UART, SPI, I2C, CAN
- **Hardware and Electronics:** Sensors, IMUs, force-sensitive resistors (FSRs), load cells, motor drivers, actuators, PCB design, KiCad, soldering, oscilloscope, logic analyzer
- **Controls and Digital Systems:** Finite-state machines (FSMs), real-time multithreading, priority inheritance/ceiling protocols, sensor fusion and redundancy, motor control, FPGA, Verilog/VHDL, fixed-point DSP, Xilinx System Generator, Simulink
- **AI/ML:** TensorFlow, Keras, Edge Impulse, INT8-quantized TensorFlow Lite (TFLite), LSTM, Gaussian-process regression, neural-network regression, CNN, RandomForest, scikit-learn, K-Means clustering, NLP
- **Industrial Automation:** DCS, PLC, SCADA, OPC, industrial Ethernet, IT/OT architecture, IP addressing, subnetting, VLANs, firewalls, OT security, Cognite Data Fusion, Power BI

ENGINEERING EXPERIENCE

Robotics Intern – Technology Innovation Institute (TII), Autonomous Robotics Research Center (ARRC)

Dec 2025 – Present

Abu Dhabi, UAE

- Developing a modular Python robotics pipeline for Forward Kinematics, Inverse Kinematics, Jacobians, workspace reachability, and singularity analysis.
- Integrated URDF parsing, Pinocchio kinematics, MuJoCo simulation, and CAD models; improved consistency between CAD geometry, coordinate frames, joint definitions, and simulation models.
- Built an interactive simulation dashboard that parses URDF parameters and updates kinematic calculations and simulation behaviour from user-controlled inputs.
- Implemented and evaluated bilinear, biquadratic, Gaussian-process, and neural-network approximation models for kinematic and workspace analysis, including error visualisation and validation.

Technology Intern – International Development Company (IDC) – Tech

Aug 2025 – Present

Industrial Automation and IT/OT Systems

- Gained hands-on exposure to industrial automation and IT/OT systems including DCS, PLC, SCADA, OPC, industrial Ethernet, IP addressing, subnetting, VLANs, firewalls, and OT security.
- Supported Factory Acceptance Test (FAT) documentation, engineering documentation, DCS server monitoring, virtual machine health checks, NTP configuration, and network/architecture activities.
- Worked with Cognite Data Fusion, Power BI, 3D asset search/browse, and Cognite InField; completed Cognite Domain Expert Basics and Cognite InField Practitioner certifications.
- Developed a C# VSTO Outlook email summarizer with AI/NLP integration and a K-Means-based file segregation workflow, iteratively debugging and tuning classification behaviour.

ENGINEERING PROJECTS

Smart Active Safety Walker – ASCEND-R (Team of 4)

Jan 2026 – Jun 2026

Embedded firmware, safety logic, sensor/actuator integration | C++, Teensy 4.1, ESP32, Edge Impulse, Gazebo

- Developed ~1,900 lines of Teensy 4.1 C++ firmware with three non-blocking control loops: 500 Hz (IMU, fall detection), 50 Hz (load cells, grip sensors, state machine), and 10 Hz (UART, battery, watchdog).
- Integrated a GY-85 IMU, 4 HX711 load cells, 2 FSR handle sensors, L298N-driven brake linear actuators, and Cytron-driven height motors, time-slicing the two height motors so only one draws current from the 12 V supply.
- Implemented a safety-oriented finite-state machine (IDLE, STANDBY, WALKING, HEIGHT_ADJUST, WARNING, EMERGENCY, FAULT) with recovery behaviour and a user-intent classifier (absent, standing, walking, leaning, unstable).
- Designed fall and safety detection: tilt thresholds, 2.5 g impact detection, 50% load-drop check, fast tip detection (tilt plus gyro rate), imbalance and grip checks, plus peak-based step and cadence detection.
- Built fail-safes including brakes engaged on power-up, battery monitoring, a hardware watchdog, and a USB calibration menu.
- Developed ESP32 firmware as a WiFi bridge serving a 5-page dashboard (overview, gait, safety, system health, settings) with WebSocket telemetry and 4-byte commands to the Teensy; height commands are rejected in EMERGENCY or FAULT.
- Deployed an INT8-quantized TensorFlow Lite model through Edge Impulse (reported accuracy 80.3%) and added browser-based voice control with an Edge Impulse keyword model (wake word "hey_ascend").

- Validated through Gazebo simulation, hardware-in-the-loop (HIL), and bench testing: ~11° warning-to-emergency threshold, ~350 ms braking window, ~2.63 s recovery, ~4.68 s height adjustment, and 80 kg support capacity.

Knee-Based Exoskeleton – Auxilium

Feb 2025 – Jun 2025

1st Place, UOWD Innovation Fair 2025 (Early Engineers Category) | Teensy 4.1, ESP32, FOC, CAN

- Developed electronics and hardware programming for a proof-of-concept knee exoskeleton designed to assist walking through controlled mechanical assistance.
- Integrated Teensy 4.1 and ESP32 controllers, IMUs, and a robotic actuator driven by Field-Oriented Control (FOC) over CAN communication.
- Led hardware integration and control implementation, refining the prototype through engineering evaluation and presentation to industry judges.

Real-Time Robotic Arm Controller

Academic Project

Real-time systems (ECTE331) | Java, multithreading

- Modelled a robotic arm controller in which SafetyMonitor (high priority), MotionPlanner (medium, 200 ms busy loop), and Logger (low) threads contend for a shared MotorController.
- Demonstrated priority inversion, then resolved it with manual priority inheritance (blocked high-priority thread boosts the lock owner) and a priority ceiling protocol, measuring SafetyMonitor wait time.

Fault-Tolerant Drone Altitude System

Academic Project

Real-time systems (ECTE331) | Java, sensor redundancy

- Simulated three redundant altitude sensors with injected faults (15% failure, 15% corrupted, 70% valid) and fused readings with 2-of-3 majority voting, requiring at least two valid readings.
- Implemented fallback to last known altitude, outlier-sensor logging, and a safe mode triggered by two consecutive reliability failures, with timestamped event logging.

Autonomous Maze Navigation – Webots e-puck Simulation

Academic Project

CSCI291 (team of 2) | C, Webots

- Programmed an e-puck robot for left-wall-following using 8 distance sensors, detecting dead-ends from repeated front-wall contact.
- Logged light intensity and GPS position at up to 10 dead-ends, identified the brightest, and stopped the robot on returning within 0.07 units of that location.

32-Band FPGA Channel Vocoder

Academic Project

ECTE433 | MATLAB, Simulink, Xilinx System Generator, Basys3 FPGA

- Designed a 32-band vocoder at 8 kHz with mel-spaced IIR bandpass filters in MATLAB; quantized coefficients to fixed-point and exported them as hex files for BRAM initialization.
- Implemented synthesis with 30 Hz low-pass envelope detection per band modulating a 140 Hz square-wave carrier, and modelled hardware quantization error in MATLAB.
- Built Simulink/System Generator hardware models (including folded and BRAM variants), converted audio to 16-bit .coe files, and verified FPGA output captured over UART (460,800 baud) against the MATLAB reference.

Space Debris Detection and Prediction – Samsung Innovation Campus

Sep 2024 – Nov 2024

Python, TensorFlow, Keras, LSTM, CNN

- Cleaned, min-max scaled, and one-hot encoded ~14,370 Space-Track SGP4 orbital records for time-series modelling.
- Built a two-layer (50-unit) LSTM over 30-step sequences of 6 orbital elements to predict next mean motion, and a 3-layer CNN regressing debris bounding boxes from 224×224 images.

SilahGrid – Intelligent Solar-Powered Edge Data Pods

Academic Project

Python, Streamlit, Plotly, scikit-learn

- Built a 1,243-line Streamlit/Plotly simulation of 10 solar-powered edge-computing pods across Dubai, modelling battery, temperature, cooling, solar input, and task queues, with offloading to a data center at low battery.
- Developed AI task routing with a RandomForest multi-output regressor (temperature, humidity, ozone) and a 200-tree RandomForest classifier selecting the target pod or data center.

AWARDS AND CERTIFICATIONS

- Best Paper Award – for published paper
- 1st Place – UOWD Innovation Fair 2025, Early Engineers Category (Auxilium knee exoskeleton)
- 3rd Place – Second Annual Inter-University Hackathon, CIHT Dubai, Feb 2025
- Dean's Merit List Award for Academic Excellence – UOWD, 2024
- AI/ML Certification – Samsung Innovation Campus, 2024
- Cognite Domain Expert Basics; Cognite InField Practitioner – Cognite, 2025

LEADERSHIP AND ACTIVITIES

UOWD Tech Club – Engineering

Jun 2024 – Sep 2025

- Engineering Head (Jul 2025 – Sep 2025); Deputy Engineering Head (Apr 2025 – Jul 2025); Engineering Executive (Jun 2024 – Apr 2025)
- Assisted in planning events, competitions, and workshops.

Engineers Australia UAE Chapter – ECT Working Member

Jul 2025 – Jul 2026

- Conducted hands-on workshops on electronics and Arduino.
- Planned and ran chapter events.

LANGUAGES

English, Telugu, Hindi, French, Arabic